

Actual Xi data and the two conductor inverse rates

Split-Zero programme: coordinate calculation

21 September 2026

This calculation continues the transport of the Alpöge–Fable Jacobian counterexample, announced by Levent Alpöge with credit to Akhil and Fable [1], through the ES four-mark extension [5] into the original weighted conductor. Section 1 gives the exact role and attribution of each construction.

Result SZ-20260921-011 contains the complete ALF1–3, AUH1–30, RXT1–9, RUC1–30 and RCM1–17 proofs. The Xi-jet residual is retained, and the tested quartet is excluded from the actual simple-zero locus in a certified neighborhood. The conductor extension there has order zero at every finite cutoff. The separate two-root geometric calculation gives uniform coefficient bounds and both inverse singular rates at the original growing Gamma cutoffs. The complete earlier proofs accompany the cumulative source. No global arithmetic zero exclusion is claimed.

Contents

1	Alpöge–Fable provenance and the exact role of the counterexample	3
2	The actual Xi-jet unit, its complete Hermite remainder and the conductor pullback	3
2.1	Two squared nodes and exact division without a zero assumption	4
2.2	The actual extended unit and all four first jets	5
2.3	The full Hermite remainder modulo the original h^2	5
2.4	The original four-dimensional primary multiplication matrix	6
2.5	The eight first-jet coordinates and the retained residual action	7
2.6	Two exact phase charts covering every nonzero actual unit	8
2.7	The exact arithmetic-jet morphism into the original conductor family	8
2.8	Every pulled-back orbit factor and the full zeroth moment	9
2.9	Exact differentials of the pullback, including all Xi and period terms	10
3	The actual Xi pullback at the certified four marks	11
3.1	Complete inputs and the exact arithmetic operations	11
3.2	Enclosures for the residual, unit and all four factors	12
3.3	The resulting operator and the excluded arithmetic neighborhood	13
4	Both original spectral poles, with coefficient bounds uniform in the cutoff	13
4.1	A single parameter neighborhood and its actual contour constants	14
4.2	Exact division with both residues and a uniform analytic remainder	14
4.3	Every original inverse coefficient, including root collision	15
4.4	The exact real phase cancellations and their limits	16
4.5	A second-order determinant identity and its explicit error	17
4.6	The finite weighted consequences, with all original constants	18
4.7	The full-matrix pole terms and their lower-triangular correction	19
4.8	The actual order-one and order-two domains	20

5	Both inverse singular rates at the original growing cutoffs	20
5.1	Two complete principal parts and the remaining analytic function	20
5.2	The exact finite rank-two matrix and its full correction	21
5.3	Both singular growth rates, including phase cancellations	22
5.4	The original amplification and the retained unit contribution	23

1 Alpöge–Fable provenance and the exact role of the counterexample

The originating Jacobian counterexample is the one announced by **Levent Alpöge** on 20 July 2026. His original announcement credits Akhil for asking the question and Fable for the work leading to the explicit polynomial [1]. The canonical ES source identifies these as Akhil Mathew and Claude Fable 5 [5]. Terence Tao subsequently gives the polynomial and its geometric explanation [10]. The counterexample and its discovery are not results of this Split-Zero calculation.

The mathematical input is precise: the three-dimensional polynomial has constant Jacobian determinant -2 , while three distinct points share one image. It supplies an explicit mechanism by which local invertibility coexists with global failure of injectivity. The ES reader then constructs the four-dimensional marked-factor extension used here, with four labelled inputs q_M, q_N, q_T, q_E [5], theorem **explicit-four-time-Jacobian-collision**. Those four inputs belong to that extension; they are not an extra preimage silently added to Alpöge’s three-dimensional fibre. Their full coordinates are retained in FC15a and FC31 of the accompanying cumulative proof.

Here is the exact use of that construction in the present programme. Let K have the four ES columns in their unchanged order, and let P be the unchanged four-dimensional polynomial of FC31. Write ℓ_a for the Lagrange polynomial at the original quartet mark ρ_a . For the original conductor T_A of actual order v , retain the spaces, columns and maps

$$\begin{aligned} \mathcal{U} &= \mathcal{P}_{v+3}/\mathcal{P}_{v-1}, & \mathcal{W} &= \mathcal{P}_3, & x_a &= [S^v \ell_a(S)], & y_a &= T_A x_a, \\ X e_a &= x_a, & Y e_a &= y_a, & \Phi &= X K^{-1}, & \Psi &= Y K^{-1}, & T_A \Phi &= \Psi. \end{aligned} \tag{ALF1}$$

FC15a proves $\det K = 77\sqrt{2}i/2 \neq 0$. FC27 proves

$$\det Y = \frac{\mu_v^4 \prod_{j=0}^3 \binom{v+j}{v}}{-64\delta^2 \gamma^2 (\delta^2 + \gamma^2)} \neq 0. \tag{ALF2}$$

These are exactly the same maps and coordinates as FC26–32; FC28–30 retains their full original Gamma metrics. Their complete proofs are in this edition’s cumulative source, preceding the new continuation. In particular, no Euclidean metric replaces those source and target norms.

Transport of the entire collision is therefore the exact identity

$$P_{\mathcal{U}} = \Phi P \Phi^{-1}, \quad P_{\mathcal{W}} = \Psi P \Psi^{-1}, \quad T_A P_{\mathcal{U}} = P_{\mathcal{W}} T_A. \tag{ALF3}$$

Indeed ALF1 gives $T_A \Phi = \Psi$ and $\Psi^{-1} T_A = \Phi^{-1}$; substituting those two equalities proves ALF3. Since $P(q_a) = (0, 0, -1, 0) = 2iq_M$, all four distinct y_a satisfy $P_{\mathcal{W}}(y_a) = 2iy_M$. This is the transported global collision. The original linear map T_A remains the exact intertwiner in ALF3. The constant-Jacobian collision is not, by itself, a zero determinant for that linear map.

The subsequent calculation follows this distinction through actual equations. FC25 computes the rank loss when the leading moment of a fixed old quotient vanishes. RPZ1–19 certifies such a geometric order-two member [8]; RLA and RQT construct its actual kernel and Gamma quotient [7, 9]. AUH and RXT below compute the Xi-jet map into the same family, retaining the residual which decides whether the marks are zeta zeros. RUC and RCM then control the inverse coefficients and their two leading growth rates. This states both the continuing role of the Alpöge–Fable input and the precise additional work done in this programme.

2 The actual Xi-jet unit, its complete Hermite remainder and the conductor pullback

The original arithmetic amplitude in PQO1–3 and OMV’s original-domain paragraph is the value of $2\xi/h$ at an actual simple quartet, not an independently chosen phase [2]. The complete

source is the preserved accepted782 cumulative LaTeX: PQO1–3, source lines 1430–1481, and OMV’s original-domain paragraph, source lines 2750–2787. The exact original period conjugation is PCL1–10 [6], with its pinned complete proof. RLA2–10 and RLA21–35 [7] give the geometric family and its exact local branches, maps and original metrics; that complete source accompanies this continuation. The following calculation constructs the pullback from the actual Xi values and derivatives and retains the nonzero-value residual. Every matrix inverse used below is displayed and checked algebraically. Hermite interpolation and the Chinese-remainder coordinates are classical constructions; their full formulas and proofs here specify the original programme objects and do not claim those constructions as new general results. The completed zeta function is Riemann’s ξ in the original theta-integral convention PQO uses, with $g = 2\xi$ retained throughout. The retained human-source account of its reflection and completion is T. M. Apostol’s DLMF 25.4.3–25.4.4 [4]; the exact theta identity used here is the one explicitly invoked by PQO3, not a new zeta identity.

2.1 Two squared nodes and exact division without a zero assumption

Let $0 < \delta < 1/2$ and $\gamma > 2$ be arbitrary real parameters in the original domain. Retain the four labels and coordinates

$$\begin{aligned} w_+ &= \delta + i\gamma, & w_- &= \delta - i\gamma, & (w_M, w_N, w_T, w_E) &= (w_+, w_-, -w_-, -w_+), & \rho_a &= 1/2 + w_a, \\ \lambda_+ &= w_+^2, & \lambda_- &= w_-^2, & d &= \lambda_+ - \lambda_- = 4i\delta\gamma \neq 0, \\ \beta &= 2(\gamma^2 - \delta^2) = -(\lambda_+ + \lambda_-), & \eta &= (\delta^2 + \gamma^2)^2 = \lambda_+\lambda_-, \\ H(y) &= (y - \lambda_+)(y - \lambda_-) = y^2 + \beta y + \eta, & h(s) &= \prod_a (s - \rho_a) = H((s - 1/2)^2). \end{aligned} \tag{AUH1}$$

The letter d here denotes the nonzero squared-node difference, not a cutoff or a conductor coefficient. The symmetry and entire-function identities from the original theta integral, recalled in PQO3, are $g(1-s) = g(s)$ and $g(\bar{s}) = \overline{g(s)}$. Consequently there is a unique entire function F with real Taylor coefficients such that

$$g(1/2 + w) = F(w^2), \quad g = 2\xi. \tag{AUH2}$$

For a proof, expand the entire left side at zero. Its odd coefficients vanish by the first symmetry and its even coefficients are real by the second. If those coefficients are c_{2n} , define $F(y) = \sum_{n \geq 0} c_{2n} y^n$. Cauchy’s coefficient bound on a circle $|w| = R$ gives $|c_{2n}| \leq M_R R^{-2n}$, so the resulting series converges for every y by taking $R^2 > |y|$. Its substitution and uniqueness follow coefficient by coefficient. No square-root branch is introduced in this definition.

Set $f_{\pm} = F(\lambda_{\pm})$ and $f'_{\pm} = F'(\lambda_{\pm})$. The exact first remainder and entire divided quotient are

$$\begin{aligned} \ell_1 &= (f_+ - f_-)/d, & \ell_0 &= (\lambda_+ f_- - \lambda_- f_+)/d, \\ L(y) &= \ell_0 + \ell_1 y = \frac{(y - \lambda_-)f_+ - (y - \lambda_+)f_-}{d}, & Q(y) &= \frac{F(y) - L(y)}{H(y)}, \\ g(s) &= L((s - 1/2)^2) + h(s)Q((s - 1/2)^2). \end{aligned} \tag{AUH3}$$

The two singularities displayed in the quotient are removable because $F - L$ vanishes at the two distinct nodes. Division of its Taylor series at each node proves removability and hence that Q is entire. It is jointly holomorphic in the distinct nodes as well: on any circle enclosing the nodes and a local y neighborhood its value is

$$Q(y) = \frac{1}{2\pi i} \int \frac{F(\zeta)}{(\zeta - \lambda_+)(\zeta - \lambda_-)(\zeta - y)} d\zeta. \tag{AUH4}$$

The residues at the two nodes and y give exactly AUH3; coefficientwise integration proves joint holomorphic dependence. Overlapping circles prove the assertion everywhere it is used. Reality

gives $\ell_0, \ell_1 \in \mathbb{R}$ and $Q(\bar{y}) = \overline{Q(y)}$ for the real original parameters. In particular the original four values of g vanish exactly when $L = 0$, equivalently $\ell_0 = \ell_1 = 0$. This equivalence follows because L has degree at most one and interpolates the two values; it does not presume that they vanish.

2.2 The actual extended unit and all four first jets

Define the two quotient values by their exact removable limits:

$$a_+ = Q(\lambda_+) = \frac{f'_+ - \ell_1}{d}, \quad a_- = Q(\lambda_-) = \frac{f'_- - \ell_1}{-d}, \quad a_- = \overline{a_+}. \quad (\text{AUH5})$$

The last equality follows by conjugating d , f'_+ and the real ℓ_1 . These are actual values of the divided function AUH3 at arbitrary original marks; they are not assigned amplitudes. They are values of the original entire unit $v_h = g/h$ precisely on the zero-residual locus $L = 0$. Away from it the exact identity is $g/h = Q + L/H$, retaining the full meromorphic residual.

The four Xi jets specify every quantity in AUH3–5:

$$\begin{aligned} f_+ &= 2\xi(\rho_M), & f_- &= 2\xi(\rho_N), \\ f'_+ &= \xi'(\rho_M)/w_+, & f'_- &= \xi'(\rho_N)/w_-, \\ a_+ &= \frac{\xi'(\rho_M)/w_+ - 2(\xi(\rho_M) - \xi(\rho_N))/d}{d}, & a_- &= \frac{\xi'(\rho_N)/w_- - 2(\xi(\rho_M) - \xi(\rho_N))/d}{-d}. \end{aligned} \quad (\text{AUH6})$$

The first line is AUH2 and the second is its derivative $g'(1/2 + w) = 2wF'(w^2)$. Neither w_+ nor w_- is zero. The map from the four scalar first-jet data to $(\ell_0, \ell_1, a_+, a_-)$ is invertible, with exact inverse

$$f_{\pm} = \ell_0 + \ell_1 \lambda_{\pm}, \quad f'_+ = \ell_1 + da_+, \quad f'_- = \ell_1 - da_-. \quad (\text{AUH7})$$

Its displayed inverse matrix has determinant $d^3 \neq 0$ in the coordinate orders

$$(\ell_0, \ell_1, a_+, a_-), \quad (f_+, f_-, f'_+, f'_-).$$

Expanding its first two rows $(1, \lambda_{\pm}, 0, 0)$ and last two rows $(0, 1, d, 0)$, $(0, 1, 0, -d)$ proves that determinant. Thus the residual and quotient-unit coordinates together retain all four original first-jet data.

At an actual simple quartet, AUH3 has $L = 0$, so $Q((s - 1/2)^2) = v_h(s)$ everywhere. The unit diagonal and its value are then exactly

$$(v_h(\rho_M), v_h(\rho_N), v_h(\rho_T), v_h(\rho_E)) = (a_+, a_-, a_-, a_+), \quad a_+ = \frac{2\xi'(\rho_M)}{h'(\rho_M)} = \frac{\xi'(\rho_M)}{4i\delta\gamma w_+} \neq 0. \quad (\text{AUH8})$$

Indeed $h'(\rho_M) = 2w_+d = 8i\delta\gamma w_+$. The nonvanishing follows exactly from simplicity; conversely, on $L = 0$ it makes all four zeros simple by reflection. This proves that the original simple-quartet data are exactly $\ell_0 = \ell_1 = 0$, $a_+ \neq 0$ in the displayed real domain. The entire function may have other zeros elsewhere; nothing in this calculation asserts a global zero-free quotient.

2.3 The full Hermite remainder modulo the original h^2

Interpolate the two values in AUH5 a second time:

$$m_1 = \frac{a_+ - a_-}{d}, \quad m_0 = \frac{\lambda_+ a_- - \lambda_- a_+}{d}, \quad M(y) = m_0 + m_1 y, \quad R_2(y) = \frac{Q(y) - M(y)}{H(y)}. \quad (\text{AUH9})$$

Both m_0, m_1 are real and R_2 is entire by the same removable division already proved. The complete identity is

$$\begin{aligned}
g(s) &= J((s-1/2)^2) + h(s)^2 R_2((s-1/2)^2), \\
J(y) &= L(y) + H(y)M(y) = j_0 + j_1 y + j_2 y^2 + j_3 y^3, \\
(j_0, j_1, j_2, j_3) &= (\ell_0 + \eta m_0, \ell_1 + \beta m_0 + \eta m_1, m_0 + \beta m_1, m_1), \\
[s^n]J((s-1/2)^2) &= \sum_{\substack{0 \leq r \leq 3 \\ 2r \geq n}} \binom{2r}{n} (-1/2)^{2r-n} j_r \quad (0 \leq n \leq 7).
\end{aligned} \tag{AUH10}$$

The last coefficient is zero when $n = 7$, by the indicated empty sum. Expansion proves every coefficient and preserves the original $1/2$ translation with its signs. The remainder is the unique polynomial of degree below eight in s matching g and g' at all four original marks. It matches because h^2 has a double zero at every mark. For uniqueness, the difference of two such polynomials is divisible by each $(s - \rho_a)^2$, hence by their product h^2 ; its degree is below eight, so it is zero. Thus no part of the Hermite data has been discarded by using the even presentation. The identities AUH7 and $a_{\pm} = m_0 + m_1 \lambda_{\pm}$ give the inverse from its coefficients to the original jets; the determinant from $(\ell_0, \ell_1, m_0, m_1)$ to those four jets is $-d^4 \neq 0$.

2.4 The original four-dimensional primary multiplication matrix

Work in the ordered centered coefficient basis $1, w, w^2, w^3$ of $A_h = \mathbb{C}[w]/(w^4 + \beta w^2 + \eta)$. Evaluation is the literal original matrix $V_{ar} = w_a^r$, with rows M, N, T, E. Its exact inverse columns and determinant are

$$(V^{-1})_{\cdot a} = \frac{1}{h'_c(w_a)} \begin{pmatrix} w_a^3 + \beta w_a \\ w_a^2 + \beta \\ w_a \\ 1 \end{pmatrix}, \quad h_c(w) = w^4 + \beta w^2 + \eta, \quad \det V = -64\delta^2 \gamma^2 (\delta^2 + \gamma^2) \neq 0. \tag{AUH11}$$

To prove the inverse, the column is the coefficient vector of $h_c(w)/((w - w_a)h'_c(w_a))$; division expands it exactly as displayed. It evaluates to one at w_a and zero at the other three marks. This proves both inverse products. The ordered Vandermonde product has factors $-2i\gamma, -2\delta, -2w_+, -2w_-, -2\delta, -2i\gamma$, giving the displayed determinant without reordering labels. Its four derivative denominators in the same order are $2w_+d, -2w_-d, 2w_-d, -2w_+d$.

The original multiplier by the entire divided quotient $Q(w^2)$ in this algebra is exactly multiplication by $M(w^2)$:

$$\begin{aligned}
\mathbf{M} &= \begin{pmatrix} m_0 & 0 & -\eta m_1 & 0 \\ 0 & m_0 & 0 & -\eta m_1 \\ m_1 & 0 & m_0 - \beta m_1 & 0 \\ 0 & m_1 & 0 & m_0 - \beta m_1 \end{pmatrix}, \quad U_{\text{ext}} = \text{diag}(a_+, a_-, a_-, a_+), \\
\mathbf{M}\mathbf{V}^{-1} &= U_{\text{ext}}, \quad \kappa = m_0^2 - \beta m_0 m_1 + \eta m_1^2 = a_+ a_- = |a_+|^2, \quad \det \mathbf{M} = \kappa^2.
\end{aligned} \tag{AUH12}$$

Reducing $w^4 = -\beta w^2 - \eta$ and $w^5 = -\beta w^3 - \eta w$ gives every matrix entry. Evaluation commutes with multiplication, proving the similarity; the product $(m_0 + m_1 \lambda_+)(m_0 + m_1 \lambda_-)$ proves κ and the determinant. On the open set $a_+ \neq 0$, its exact inverse is obtained from the same displayed matrix with

$$(m_0, m_1) \mapsto (n_0, n_1) = \left(\frac{m_0 - \beta m_1}{\kappa}, -\frac{m_1}{\kappa} \right), \quad U_{\text{ext}}^{-1} = \text{diag}(a_+^{-1}, a_-^{-1}, a_-^{-1}, a_+^{-1}). \tag{AUH13}$$

In fact $(m_0 + m_1 y)((m_0 - \beta m_1) - m_1 y) = \kappa - m_1^2 H(y)$, which proves the inverse in the quotient, and hence both matrix inverse products.

For the original uncentered coefficient basis $1, s, s^2, s^3$, retain $T_{rb} = \binom{b}{r} 2^{-(b-r)}$ for $r \leq b$, zero otherwise. It has determinant one, inverse $T_{rb}^{-1} = \binom{b}{r} (-1/2)^{b-r}$, and the matrices are exactly

$$V_\rho = VT, \quad V_\rho^{-1} = T^{-1}V^{-1}, \quad M_s = T^{-1}MT, \quad M_s^{-1} = T^{-1}M^{-1}T. \quad (\text{AUH14})$$

Substitution $s = 1/2 + w$ proves the two translation matrices compose to the identity and establishes these formulas. These are coordinate morphisms, not isometries. For example if $H_{s_0,c}$ is the original Gamma Gram of the uncentered monomials, the centered coefficient metric is $(T^{-1})^* H_{s_0,c} T^{-1}$ and the primary evaluation metric is $(V_\rho^{-1})^* H_{s_0,c} V_\rho^{-1}$. Here $s_0 \in \{1, a_{\text{amp}}\}$ distinguishes the original Gamma parameter from the polynomial variable. Its mass, center and all cross terms remain the original ones of WCF2 and RLA35; no amplitude or metric factor has been set to one.

2.5 The eight first-jet coordinates and the retained residual action

Every polynomial class modulo h_c^2 has a unique expression

$$P_0(w) + h_c(w)P_1(w), \quad \deg P_0, \deg P_1 < 4.$$

Successive division by the monic quartic proves existence and uniqueness. Define $\mathbf{L}$ by AUH12's displayed matrix with (m_0, m_1) replaced by (ℓ_0, ℓ_1) , and let $(\mathbf{K}_L)_{0,2} = (\mathbf{K}_L)_{1,3} = \ell_1$ be its only nonzero entries. The full multiplier by g in these eight coordinates is

$$\mathbf{G}_8 = \begin{pmatrix} \mathbf{L} & 0 \\ \mathbf{M} + \mathbf{K}_L & \mathbf{L} \end{pmatrix}. \quad (\text{AUH15})$$

Indeed multiplication of $L(w^2)$ by P_0 has reduced part $\mathbf{L}P_0$ and the exact carry $h_c \ell_1((P_0)_2 + (P_0)_3 w)$. Multiplication of $h_c M(w^2)$ by P_0 , and of $h_c L(w^2)$ by P_1 , gives the remaining lower block; every further carry has a factor h_c^2 and is zero in the stated quotient. This proves AUH15 from AUH10, including the residual carry which would be lost by replacing the nonzero residual by a unit.

Let $V'_{ar} = r w_a^{r-1}$, with its column at $r = 0$ equal to zero, and let $D_h = \text{diag}(h'_c(w_a))$ in the unchanged order. The exact evaluation-and-derivative map, and its inverse, are

$$\begin{aligned} \mathbf{J}_8 &= \begin{pmatrix} V & 0 \\ V' & D_h V \end{pmatrix}, & \begin{pmatrix} \mathbf{v} \\ \mathbf{j} \end{pmatrix} &= \mathbf{J}_8 \begin{pmatrix} P_0 \\ P_1 \end{pmatrix}, \\ P_0 &= V^{-1} \mathbf{v}, & P_1 &= V^{-1} D_h^{-1} (\mathbf{j} - V' V^{-1} \mathbf{v}), \\ \mathbf{J}_8 \mathbf{G}_8 \mathbf{J}_8^{-1} &= \begin{pmatrix} D_g & 0 \\ D_{g'} & D_g \end{pmatrix}, & (D_g)_{aa} &= L(w_a^2), \quad (D_{g'})_{aa} = 2w_a \ell_1 + h'_c(w_a) a_a, \\ (a_M, a_N, a_T, a_E) &= (a_+, a_-, a_-, a_+). \end{aligned} \quad (\text{AUH16})$$

The first line is the derivative of $P_0 + h_c P_1$ at each root; substitution verifies both inverse compositions. The third line is the product rule $(gP)' = g'P + gP'$, with g, g' from AUH10. It proves the full eight-jet identity, not just its four-value restriction. If both sampled values f_+, f_- are nonzero, then $\mathbf{L}$ is inverted by AUH13 with the ℓ coefficients, and the entire eight-dimensional inverse is

$$\mathbf{G}_8^{-1} = \begin{pmatrix} \mathbf{L}^{-1} & 0 \\ -\mathbf{L}^{-1}(\mathbf{M} + \mathbf{K}_L)\mathbf{L}^{-1} & \mathbf{L}^{-1} \end{pmatrix}. \quad (\text{AUH17})$$

At the original simple quartet, instead, $L = 0$ and AUH15 is $\begin{pmatrix} 0 & 0 \\ \mathbf{M} & 0 \end{pmatrix}$, with square zero and rank four, since $\mathbf{M}$ is invertible. Thus the unit and the nilpotent multiplication by g have their different exact maps retained. Formula AUH16 remains valid at every stratum, whether or not AUH17 is invertible.

2.6 Two exact phase charts covering every nonzero actual unit

Put $r = (a_+ + a_-)/2 = \Re a_+$ and $c = (a_+ - a_-)/(2i) = \Im a_+$. They are determined by AUH6, not free coordinates in the pullback. Write $J_d = \text{diag}(1, -1, -1, 1)$, so $U_{\text{ext}} = rI + icJ_d$. The complete two-chart dictionary and its inverses are

chart	coordinates	U_{ext}	matrix inverse
$r \neq 0$	$(r, x = c/r)$	$rW_R(x), \quad W_R = I + ixJ_d$	$W_R^{-1} = (I - ixJ_d)/(1 + x^2)$
$c \neq 0$	$(c, y = r/c)$	$cW_I(y), \quad W_I = yI + iJ_d$	$W_I^{-1} = (yI - iJ_d)/(1 + y^2)$

(AUH18)

The inverse coordinate formulas are $a_+ = r(1 + ix)$ and $a_+ = c(y + i)$, with a_- its conjugate. On the overlap,

$$y = 1/x, \quad c = rx, \quad r = cy, \quad W_I(1/x) = x^{-1}W_R(x). \quad (\text{AUH19})$$

Direct multiplication proves the matrix inverses and transitions. Since $a_+ \neq 0$ exactly when $(r, c) \neq (0, 0)$, these two charts cover every nonzero actual unit, including a purely imaginary one. The original unit magnitude is $r^2 + c^2 = |a_+|^2 = \kappa$, not a chosen unit modulus. It cancels only where an operator contains U_{ext}^{-1} and U_{ext} as a conjugate pair. All period matrices and original metrics still retain the full U_{ext} .

2.7 The exact arithmetic-jet morphism into the original conductor family

For the original nonzero period u on its unchanged logarithm branch, put $z = u^{-1}$, $\mathfrak{a} = 1/160 + \beta/24 + \eta/2$, $\zeta = e^{2\pi i/5}$ and $D_j = \text{diag}(\zeta^{-ja})_{a=1}^4$. Use the entire original period matrix, with every coefficient,

$$\begin{aligned} \mathcal{R}_{ar}(z; \delta, \gamma) &= \sum_{n \geq 0} \sum_{\substack{p, h \geq 0 \\ 2p+4h=5n+r+1-a}} \frac{(\beta/3)^p \eta^h (-5)^{p+h-n} (a/5)_{p+h-n}}{p!h!} z^n, \\ \mathbf{H}_{\text{ext}}(u) &= e^{a/u} G_u \mathcal{R}(u^{-1}; \delta, \gamma) V^{-1} U_{\text{ext}}, \\ G_u &= \text{diag}(u^{a/5} \mathfrak{g}_a)_{a=1}^4, \quad \mathfrak{g}_a = 5^{a/5-1} e^{\pi i a/5} \Gamma(a/5), \\ \mathbf{C}_j &= V \mathcal{R}^{-1} D_j \mathcal{R} V^{-1}, \quad \mathbf{B}_j = U_{\text{ext}}^{-1} \mathbf{C}_j U_{\text{ext}}. \end{aligned} \quad (\text{AUH20})$$

The index ranges are $1 \leq a \leq 4$, $0 \leq r \leq 3$, and empty inner sums are zero. PCL1–5 prove convergence and $\det \mathcal{R} = 1$, with its exact inverse $\text{adj } \mathcal{R}$. PQO8–9 and PCL3 specify the retained Gamma ray factors $\mathfrak{g}_a$; every power of u in G_u uses the same original branch. The original conjugation identity is exactly

$$\mathbf{H}_{\text{ext}}^{-1} D_j \mathbf{H}_{\text{ext}} = \mathbf{B}_j. \quad (\text{AUH21})$$

Indeed the nonzero scalar exponential cancels with its inverse and G_u commutes with D_j . The inverse of the entire left-hand coordinate map, where $a_+ \neq 0$, is explicitly $U_{\text{ext}}^{-1} V \mathcal{R}^{-1} G_u^{-1} e^{-a/u}$. No Gamma factor or center is removed from that map.

Define the source parameter domain $\mathcal{D} = \{(\delta, \gamma, u) : 0 < \delta < 1/2, \gamma > 2, u \neq 0, a_+(\delta, \gamma) \neq 0\}$, where a_+ is the actual Xi-jet expression AUH6. The following are exact morphisms into the extended original geometric family, with the residual carried as additional output:

$$\begin{aligned} \Psi : (\delta, \gamma, u) &\longmapsto (\delta, \gamma, z = u^{-1}, a_+, a_-; \ell_0, \ell_1), \\ \Psi_R : (\delta, \gamma, u) &\longmapsto (\delta, \gamma, z = u^{-1}, r, x = c/r; \ell_0, \ell_1) \quad (r \neq 0), \\ \Psi_I : (\delta, \gamma, u) &\longmapsto (\delta, \gamma, z = u^{-1}, c, y = r/c; \ell_0, \ell_1) \quad (c \neq 0). \end{aligned} \quad (\text{AUH22})$$

Their inverse onto their respective graphs retains δ, γ and takes $u = 1/z$; AUH18–19 recover every unit and residual coordinate. Thus these are not freely assigned phase families. The first

chart uses exactly RLA2's $W(x)$; the second gives its regular complementary chart by the proved transition. The quartet parameters in AUH20 are retained as variables; fixing them at $(1/4, 3)$ gives precisely RLA2's coefficient family. The isomorphism of local branches proved at the RLA base point is not asserted at every image of AUH22. AUH22 itself is an exact map on its entire stated domain, requiring no assumed zeros.

The actual simple-quartet locus in this domain is exactly $\mathcal{Z} = \{\ell_0 = \ell_1 = 0\}$, by AUH3 and AUH8. On its complement AUH22 still composes the original geometric matrix formula with the actual divided Xi-jet values, and retains the exact discrepancy L/h from the original quotient. Consequently the map does not label that nonzero-residual extension as an arithmetic simple-quartet unit. This distinction is given by the explicit equality AUH3, not by an unconstructed comparison.

2.8 Every pulled-back orbit factor and the full zeroth moment

To distinguish the original Riemann function from the exponential spectral coordinate, write that latter coordinate as τ in this subsection; it is exactly the variable called ξ in RLA2. With $e(\tau) = (e^{\rho_a \tau})_a$, the complete composition is

$$\begin{aligned} f_j^\Xi(\tau; \delta, \gamma, u) &= Q_0(B_j e(\tau)), & Q_0(v) &= v_1 v_4 - v_2 v_3, \\ E_A^\Xi(\tau; \delta, \gamma, u) &= \prod_{j=1}^4 f_j^\Xi(\tau), & \mu_n^\Xi &= \partial_\tau^n E_A^\Xi(0), \\ g_{\text{cond}}^\Xi(t) &= e^{-4\omega(t)} E_A^\Xi(\omega(t)), & \omega(t) &= -i \arctan t. \end{aligned} \tag{AUH23}$$

Every entry of B_j is the convergent matrix expression AUH20 with the actual Xi samples AUH6. It is therefore an explicit composition, with every original ordered-pair multiplicity retained by the products. The original symbol in WCF and RLA, its -4 center correction, every moment and the unchanged Gamma weights are all pulled back by this identity.

There is also an exact finite expression displaying the amplitude dependence without concealing the residual constraint. Let $e_+ = (e^{\rho_M \tau}, 0, 0, e^{\rho_E \tau})^\top$, $e_- = (0, e^{\rho_N \tau}, e^{\rho_T \tau}, 0)^\top$, and write $p_{jr} = (C_j e_+)_r$, $q_{jr} = (C_j e_-)_r$. For $\vartheta = a_-/a_+$, the full factor is

$$\begin{aligned} f_j^\Xi &= p_{j1} p_{j4} + \vartheta(p_{j1} q_{j4} + q_{j1} p_{j4}) + \vartheta^2 q_{j1} q_{j4} \\ &\quad - \vartheta^{-2} p_{j2} p_{j3} - \vartheta^{-1}(p_{j2} q_{j3} + q_{j2} p_{j3}) - q_{j2} q_{j3}. \end{aligned} \tag{AUH24}$$

This follows by applying U_{ext} to $e_+ + e_-$ and U_{ext}^{-1} to the four output coordinates in AUH20; in particular its four signs and its two cross terms are exact. The common amplitude scale cancels here because it occurs in a conjugate matrix pair; its value remains in AUH20's coordinate map. It has not been assigned by selecting ϑ .

An expression without any phase chart is

$$\begin{aligned} \mathcal{D}_j(\tau) &= a_-^2(a_+ p_{j1} + a_- q_{j1})(a_+ p_{j4} + a_- q_{j4}) \\ &\quad - a_+^2(a_+ p_{j2} + a_- q_{j2})(a_+ p_{j3} + a_- q_{j3}), \\ f_j^\Xi(\tau) &= \frac{\mathcal{D}_j(\tau)}{a_+^2 a_-^2}, & \mu_0^\Xi(\delta, \gamma, u) &= \frac{\prod_{j=1}^4 \mathcal{D}_j(0)}{(a_+ a_-)^8}. \end{aligned} \tag{AUH25}$$

Both equalities follow by clearing exactly the displayed denominators in AUH24. The numerators remain defined even at vanishing quotient values; the inverse coordinate map at such a value is not asserted. For real $u \neq 0$, the original reflection calculation RLA2 gives $f_{5-j}^\Xi(\tau) = -\overline{f_j^\Xi(\bar{\tau})}$. Indeed the coefficient series is real, $\bar{V} = J_c V$, $\bar{U}_{\text{ext}} = J_c U_{\text{ext}} J_c$ and $Q_0(J_c v) = -Q_0(v)$, exactly as in the original proof. Therefore the exact real-parameter pullback is

$$\mu_0^\Xi = |f_1^\Xi(0) f_2^\Xi(0)|^2 = \frac{|\mathcal{D}_1(0) \mathcal{D}_2(0)|^2}{(a_+ a_-)^8}. \tag{AUH26}$$

This is the complete zeroth moment on the actual Xi-jet graph. Its zero set there is exactly the union of the two explicit numerator zero sets. The arithmetic simple-quartet intersection also retains the two explicit equations $\ell_0 = \ell_1 = 0$. These are pullbacks of the actual functions, rather than a claim that an arbitrary geometric phase pair meets that intersection.

2.9 Exact differentials of the pullback, including all Xi and period terms

To make the parameter morphism usable in local calculations, retain ordinary differentials in the original variables:

$$\begin{aligned}
dw_{\pm} &= d\delta \pm i d\gamma, & d\lambda_{\pm} &= 2w_{\pm}dw_{\pm}, \\
d(d) &= 4i(\gamma d\delta + \delta d\gamma), \\
d\ell_1 &= \frac{f'_+ d\lambda_+ - f'_- d\lambda_- - \ell_1 d(d)}{d}, \\
d\ell_0 &= df_+ - \lambda_+ d\ell_1 - \ell_1 d\lambda_+, \\
da_+ &= \frac{F''(\lambda_+)d\lambda_+ - d\ell_1 - a_+ d(d)}{d}, \\
da_- &= \frac{F''(\lambda_-)d\lambda_- - d\ell_1 + a_- d(d)}{-d}, \\
F''(\lambda_{\pm}) &= \frac{\xi''(\rho_{M/N})}{2w_{\pm}^2} - \frac{\xi'(\rho_{M/N})}{2w_{\pm}^3}, \\
dz &= -u^{-2}du.
\end{aligned} \tag{AUH27}$$

Here $d(d)$ explicitly means the differential of the scalar $d = 4i\delta\gamma$; $df_+ = f'_+ d\lambda_+$. Each identity is differentiation of AUH3–6; the second derivative formula follows by differentiating $g'(1/2 + w) = 2wF'(w^2)$. Thus every second Xi derivative needed for the actual phase pullback is specified. The exact chart derivatives are

$$dr = (da_+ + da_-)/2, \quad dc = (da_+ - da_-)/(2i), \quad dx = (r dc - c dr)/r^2, \quad dy = (c dr - r dc)/c^2. \tag{AUH28}$$

They apply on the respective charts of AUH18, and retain the actual scale derivative rather than holding a free phase fixed.

Finally let $K = \mathcal{R}V^{-1}U_{\text{ext}}$, so that $B_j = K^{-1}D_jK$. Every parameter derivative of the pulled-back original matrices is

$$\begin{aligned}
\mathcal{A} &= K^{-1}dK = U_{\text{ext}}^{-1}V\mathcal{R}^{-1}(d\mathcal{R})V^{-1}U_{\text{ext}} - U_{\text{ext}}^{-1}(dV)V^{-1}U_{\text{ext}} + U_{\text{ext}}^{-1}dU_{\text{ext}}, \\
dB_j &= [B_j, \mathcal{A}], & (dV)_{ar} &= rw_a^{r-1}dw_a, \\
d\mathcal{R} &= \mathcal{R}_z dz + \mathcal{R}_{\beta} d\beta + \mathcal{R}_{\eta} d\eta, \\
d\beta &= 4\gamma d\gamma - 4\delta d\delta, & d\eta &= 4(\delta^2 + \gamma^2)(\delta d\delta + \gamma d\gamma), \\
dU_{\text{ext}} &= \text{diag}(da_+, da_-, da_-, da_+).
\end{aligned} \tag{AUH29}$$

The coefficients of $\mathcal{R}_{\beta}, \mathcal{R}_{\eta}$ are the literal term derivatives of AUH20: a term indexed by (p, h) is multiplied respectively by p/β and h/η , with zero for $p = 0$ or $h = 0$. Both β, η are positive on the real original domain. Termwise differentiation is justified on compact parameter sets by the original factorially convergent period series, or by its absolutely convergent period integral with the fifth-degree decay and polynomial parameter derivatives. For an explicit majorant, choose a common constant $K_0 \geq 1$ on the compact set with $K_0^2 \geq 5|\beta|/3$, $K_0^4 \geq 5|\eta|$. The coefficient argument of RPZ5 [8] bounds the entire coefficient by $4K_0^3(2n+2)(6K_0^5/5)^n/n!$. On a slightly larger compact set with nonzero β, η the first coefficient derivatives introduce factors at most linear in n , because $p, h \leq (5n+3)/2$; higher fixed derivatives introduce fixed polynomial factors. The resulting series still converge uniformly. This supplies the differentiation used in AUH29. Differentiation of $K^{-1}D_jK$ proves the commutator, including its order and all three terms in $\mathcal{A}$.

Let $v_j = B_j e(\tau)$ and retain the original bilinear polar form $\mathfrak{b}(v, w) = v_1 w_4 + v_4 w_1 - v_2 w_3 - v_3 w_2$. Then the full first variation is

$$\begin{aligned} df_j^\Xi &= \mathfrak{b}(v_j, [B_j, \mathcal{A}]e(\tau) + B_j de(\tau)), \\ (de(\tau))_a &= e^{\rho_a \tau} (\rho_a d\tau + \tau d\rho_a), \\ d\mu_0^\Xi &= \sum_{j=1}^4 df_j^\Xi(0) \prod_{k \neq j} f_k^\Xi(0). \end{aligned} \tag{AUH30}$$

The first line follows from $Q_0(v) = \mathfrak{b}(v, v)/2$ and symmetry of this bilinear form, with no Hermitian substitution. The last line is the exact product rule, valid also at its zero set without division by a vanishing factor. Equations AUH27–30 give the actual tangent pullback into RLA’s factor and moment equations, including changes of the quartet, unit, period and original spectral coordinate.

This completes the morphism from the actual Xi-jet data to the original conductor and its local algebra wherever the latter’s coordinate chart is used. The full residual, the unit scale and both phase charts remain explicit. The calculation proves the map and its identities; it does not assert that the previously certified geometric singular point is on this arithmetic graph.

3 The actual Xi pullback at the certified four marks

AUH1–30 constructs the exact map from the original Xi values and derivatives to the full period family, carrying the zero-value residual. This section evaluates that map on the quartet used in RPZ1–19 [8], and on its entire certified real-period interval. It does not substitute an assigned phase for the actual quotient values. The resulting map is the extension defined by AUH3 and AUH20–26; the residual below determines whether it is on the original simple-zero-quartet locus.

The four labels retain the exact programme transport FC1–55 of the four-time construction in the canonical ES reader by The Clankers, theorem `explicit-four-time-Jacobian-collision` [5]. The complete original source and its proof locator accompany this edition.

3.1 Complete inputs and the exact arithmetic operations

Retain all four ordered marks

$$(\rho_M, \rho_N, \rho_T, \rho_E) = (3/4 + 3i, 3/4 - 3i, 1/4 + 3i, 1/4 - 3i), \quad \delta = 1/4, \quad \gamma = 3. \tag{RXT1}$$

Let z_c and x_c be the exact decimal rational centers

$$\begin{aligned} z_c &= 0.02211947780866184513200791639903555836071244787217846841, \\ x_c &= -1.466817440463494183975448871845763045856183562396293859, \\ I_z &= [z_c - 10^{-15}, z_c + 10^{-15}], \quad I_x = [x_c - 10^{-15}, x_c + 10^{-15}]. \end{aligned} \tag{RXT2}$$

RPZ proves a unique geometric pair $(z_*, x_*) \in I_z \times I_x$. Here $u = 1/z$ remains the original positive real period, for every $z \in I_z$. The Xi-derived phase below depends on the quartet, not on the period, so its evaluation applies at z_* as well.

Put $s_0 = 3/4 + 3i$ and $w_+ = 1/4 + 3i$. The completed function is Riemann’s original one, in the convention $g = 2\xi$ of PQO1 [2, 4]. The certificate evaluates its Taylor series at s_0 through degree two by the exact formula

$$X(\varepsilon) = \frac{1}{2}(s_0 + \varepsilon)(s_0 + \varepsilon - 1) \exp\left(-\frac{s_0 + \varepsilon}{2} \log \pi\right) \Gamma\left(\frac{s_0 + \varepsilon}{2}\right) \zeta(s_0 + \varepsilon) \pmod{\varepsilon^3}. \tag{RXT3}$$

Thus $X_0 = \xi(s_0)$, $X_1 = \xi'(s_0)$, and $2X_2 = \xi''(s_0)$. Every factor is holomorphic on a neighborhood of the input rectangles used here. This is an exact derivative calculation in a truncated power-series algebra, not a truncation of the original period function.

For real δ, γ , AUH3–6 reduce to the following exact real arithmetic, with $w_+ = \delta + i\gamma$ and X_j evaluated at $1/2 + w_+$:

$$\begin{aligned} \ell_1 &= \frac{\Im(2X_0)}{2\delta\gamma}, & \ell_0 &= \Re(2X_0) - \ell_1(\delta^2 - \gamma^2), \\ a_+ &= \frac{X_1/w_+ - \ell_1}{4i\delta\gamma}, & x_\Xi &= \frac{\Im a_+}{\Re a_+} \quad (\Re a_+ \neq 0). \end{aligned} \tag{RXT4}$$

Indeed $f_- = \overline{f_+}$, $f_+ = 2X_0$, and $\lambda_+ - \lambda_- = 4i\delta\gamma$. Substitution into AUH3 and AUH5 gives the four identities with every factor of two retained. AUH8 explains precisely when a_+ is the original simple-quartet unit: the additional two residual coordinates must vanish. RXT4 does not impose that equality.

3.2 Enclosures for the residual, unit and all four factors

The accompanying executable `certify_actual_xi_pullback.py` uses python-flint 0.9.0, Arb arithmetic at 768 bits and one worker. Johansson’s original account [12], in its discussion of zeta-series methods and its sections “Precision and bounds” and “Polynomials, power series and matrices”, supplies the interval inclusion contract and power-series arithmetic used here. All interval inputs are exact decimal rationals enlarged by outward-rounded balls. The receipt preserves the complete computed enclosures and source hashes. In particular it proves the rational bounds

$$\begin{aligned} 0.97625 < \ell_0 < 0.97627, & \quad 0.01893 < \ell_1 < 0.01894, & \quad \Re a_+ > 0, \\ 0.01020 < x_\Xi < 0.01022, & \quad x_\Xi \notin I_x. \end{aligned} \tag{RXT5}$$

For readable values, the same enclosure gives

$$\begin{aligned} a_+ &\simeq 0.00020586446904694959 + 0.000002101993022349772i, \\ x_\Xi &\simeq 0.010210567331414483. \end{aligned}$$

These displayed approximations are not used as exact inputs. The inequalities in RXT5 are comparisons performed on enclosing balls, including their radii, rather than on rounded mid-points.

The certificate next evaluates the complete original period matrix PCL2 at every $z \in I_z$, with the Xi-derived phase enclosed by RXT4. It retains terms $0 \leq n \leq 650$ and adds the entire three tail bounds proved in RPZ5–8. The coefficient kernel, Taylor matrix enclosure and derivatives are copied without changes from the previously certified script whose SHA256 is `bb50ba841052d305fc1a66bbfab9095af816f4c8b05eb3f5a1fa578dab883bbe`. The new calculation replaces the phase input by the actual value in RXT4. The full unit scale cancels only in the conjugate pair AUH21, whose complete coordinate map retains that scale.

The four values $d_j^\Xi = f_j^\Xi(0)$ are enclosed as follows; each displayed radius is part of the inclusion:

j	$\Re d_j^\Xi$	$\Im d_j^\Xi$	
1	$0.27718630 \pm 4.03 \cdot 10^{-9}$	$-0.382818517 \pm 7.47 \cdot 10^{-10}$	
2	$-0.121851989 \pm 9.13 \cdot 10^{-10}$	$0.134377151 \pm 7.86 \cdot 10^{-10}$	
3	$0.121851989 \pm 9.13 \cdot 10^{-10}$	$0.134377151 \pm 7.86 \cdot 10^{-10}$	
4	$-0.27718630 \pm 4.03 \cdot 10^{-9}$	$-0.382818517 \pm 7.47 \cdot 10^{-10}$	

(RXT6)

The exact reflection $d_{5-j}^\Xi = -\overline{d_j^\Xi}$ is AUH26; it is not inferred from this rounded table. Therefore the original zeroth-moment formula, evaluated on the extension, satisfies throughout I_z

$$\mu_0^\Xi = \prod_{j=1}^4 d_j^\Xi = |d_1^\Xi d_2^\Xi|^2, \quad 0.0073504212 < \mu_0^\Xi < 0.0073504216. \tag{RXT7}$$

Both comparisons and all four nonvanishing tests are executable assertions in the certificate. Since the period matrix includes the proved analytic tails, RXT6–7 enclose the original entire series rather than a polynomial approximation to it.

3.3 The resulting operator and the excluded arithmetic neighborhood

For every finite $D \geq 0$ and original $s \in \{1, a_{\text{amp}}\}$, put $\rho_j = \sqrt{j!/(s/2)_j}$ and retain the original Gamma source and target centers $a_{\text{amp}}/2$ and $a_{\text{amp}}/2 - 4$, and the mass $(2\pi)^{s/2}$. Write $g_n^\Xi = [t^n]g_{\text{cond}}^\Xi(t)$ using the complete AUH23 symbol. Then the extended original conductor has order zero throughout I_z , and its exact matrix and inverse are

$$\begin{aligned} B_{ij}^\Xi &= \mathbf{1}_{j \geq i} g_{j-i}^\Xi \rho_j / \rho_i, & \det B^\Xi &= (\mu_0^\Xi)^{D+1} \neq 0, \\ h_0^\Xi &= (\mu_0^\Xi)^{-1}, & h_n^\Xi &= -(\mu_0^\Xi)^{-1} \sum_{j=1}^n g_j^\Xi h_{n-j}^\Xi, & (B^\Xi)_{ij}^{-1} &= \mathbf{1}_{j \geq i} h_{j-i}^\Xi \rho_j / \rho_i. \end{aligned} \quad (\text{RXT8})$$

Multiplication of the two triangular matrices reduces entrywise to $\sum_{j=0}^n g_j^\Xi h_{n-j}^\Xi = \mathbf{1}_{n=0}$, proving both inverse products. Triangularity proves the determinant; the diagonal Gamma factors cancel in that determinant, with neither metric removed. Thus the singular old-domain example at x_* does not give a singular matrix after the Xi-jet substitution at these same four marks and the same original period interval.

There is also a certified open neighborhood in the quartet parameters. Evaluate RXT3–4 by the same inclusion arithmetic on the complete real rectangle

$$|\delta - 1/4| \leq 10^{-8}, \quad |\gamma - 3| \leq 10^{-8}. \quad (\text{RXT9})$$

The receipt proves $\ell_0 > 0.975$, $\ell_1 > 0.0188$, $\Re a_+ > 0$ and $x_\Xi > 0$ on the whole rectangle. No sample-grid inference is made. AUH3 proves that an actual simple zero quartet requires $\ell_0 = \ell_1 = 0$; hence this rectangle contains none, for any original period. This is a local arithmetic exclusion with a stated domain. It does not exclude quartets at other heights, nor assert that the order-zero extension away from the residual-zero locus is the original arithmetic quotient $2\xi/h$.

The exact surviving comparison is AUH3: $2\xi/h = Q + L/H$. RXT5 and RXT9 evaluate its retained residual, while RXT7–8 evaluate the complete conductor constructed from Q . Both sides and their discrepancy are explicit. The geometric counterexample and the arithmetic pullback are therefore compared by their actual map, rather than by replacing either one with the other. RUC and RCM continue to describe the nearby geometric two-root inverse; RXT determines what the tested arithmetic graph does at the specific certified marks.

4 Both original spectral poles, with coefficient bounds uniform in the cutoff

The functions in this section are the actual functions of RLA1–10, on the neighborhood constructed there from the certified point RPZ1–19 [8]. Their complete sources accompany this derivation. The four labels M, N, T, E and primary exponents in RLA1 are unchanged. In particular we retain the original $p = (z, x)$, $u = 1/z$, unit $a = \alpha_{\text{unit}}(1 + ix)$, spectral coordinate t , and $\omega(t) = -i \arctan t$ throughout. The exact maps $p = \mathcal{P}_{\text{roots}}(\alpha, \beta)$ in RLA9 recover both original parameters whenever root coordinates are used below. This continuation concerns that same geometric family; it makes no identification with an arithmetic zeta quartet.

The complete factorization proved in RLA5–10 is

$$\begin{aligned}
g(t, p) &= e^{-4\omega(t)} \prod_{j=1}^4 f_j(\omega(t), p) = U(t, p)(t - \alpha(p))(t - \beta(p)), & F(t, p) &= U(t, p)^{-1}, \\
U &= e^{-4\omega} U_+ U_-, & U(0, p_*) &= -b, \quad b = |G_\xi(0, p_*)|^2 > 0, & G &= f_1 f_2, \\
U_+(t, p) &= \int_0^1 \partial_t [G(\omega(\cdot), p)] (\alpha + \theta(t - \alpha)) d\theta,
\end{aligned} \tag{RUC1}$$

and U_- is the same integral for $G^\#(\omega(\cdot), p)$ at the root β . The original center exponential and all unit coefficients remain in U . The integral divisions and the existence of the two roots are proved in RLA5–6 by contraction and integration of the complete analytic functions.

4.1 A single parameter neighborhood and its actual contour constants

There is a positive dyadic $R < 1/2$ such that, for $r_0 = R/4$, the full functions and the inverse root map are holomorphic on a neighborhood of

$$\{|t| \leq 2R, |\alpha| \leq r_0, |\beta| \leq r_0\}, \quad |U(t, \mathcal{P}_{\text{roots}}(\alpha, \beta)) + b| \leq b/2. \tag{RUC2}$$

To verify existence, RLA9 first gives an open neighborhood of the origin in the two root coordinates. Composing its holomorphic inverse with RLA6 gives a holomorphic function of (t, α, β) equal to $-b$ at the origin. Continuity supplies a polydisc on which the displayed difference is strictly less than $b/2$. Every sufficiently small dyadic R has its displayed closed polydisc inside that open set. Fix any one such R once and for all. This is a proved choice of a neighborhood of the actual functions; it is not an extra nonvanishing hypothesis on freely assigned coefficients.

Let $\mathcal{K} = \mathcal{P}_{\text{roots}}(\{|\alpha|, |\beta| \leq r_0\})$. It is a compact neighborhood of the point in its original parameter space. Define the actual contour maximum and the following completely specified constants:

$$\begin{aligned}
M &= \max_{p \in \mathcal{K}, |\zeta|=R} |F(\zeta, p)| \leq 2/b, & m &= 2/(3b), & |F(t, p)| &\geq m \quad (|t| \leq 2R), \\
C_0 &= \frac{MR}{R - r_0}, & C_1 &= \frac{MR}{(R - r_0)^2}, & C &= \frac{M}{(R - r_0)^2}, & A &= C_0 + 2r_0 C_1.
\end{aligned} \tag{RUC3}$$

The inequalities for F follow directly from $b/2 \leq |U| \leq 3b/2$. No parameter or cutoff dependence is hidden in M : it is the maximum of the stated actual function on the specified fixed compact contour family. The constants C_0, C_1 bound respectively $|F(a, p)|$ and $|\partial_t F(a, p)|$ for $|a| \leq r_0$. Indeed Cauchy's integral formula and its derivative give denominators $\zeta - a$ and $(\zeta - a)^2$, respectively; the contour length is $2\pi R$ and their moduli are at least $R - r_0$ and $(R - r_0)^2$.

4.2 Exact division with both residues and a uniform analytic remainder

For $\alpha \neq \beta$ define the degree-one interpolation polynomial

$$\begin{aligned}
L(t, p) &= \frac{F(\alpha, p)(t - \beta) - F(\beta, p)(t - \alpha)}{\alpha - \beta} = \ell_0 + \ell_1 t, \\
\ell_1 &= \frac{F(\alpha, p) - F(\beta, p)}{\alpha - \beta}, & \ell_0 &= F(\alpha, p) - \alpha \ell_1.
\end{aligned} \tag{RUC4}$$

At $\alpha = \beta = a$ use exactly $L(t, p) = F(a, p) + \partial_t F(a, p)(t - a)$. All these functions extend holomorphically across the diagonal: the divided difference in RUC4 equals $\int_0^1 \partial_t F(\beta + \theta(\alpha - \beta), p) d\theta$. Every segment lies in $|t| \leq r_0$. Thus $|\ell_1| \leq C_1$ and $|\ell_0| \leq C_0 + r_0 C_1$, uniformly on $\mathcal{K}$.

The exact decomposition, on the entire neighborhood, is

$$\begin{aligned} \frac{1}{g(t, p)} &= \frac{L(t, p)}{(t - \alpha)(t - \beta)} + H(t, p), \\ H(t, p) &= \frac{1}{2\pi i} \int_{|\zeta|=R} \frac{F(\zeta, p)}{(\zeta - \alpha)(\zeta - \beta)(\zeta - t)} d\zeta \quad (|t| < R). \end{aligned} \quad (\text{RUC5})$$

The equality of meromorphic functions includes parameters where a root is zero. To prove it first take t, α, β distinct. Partial fractions in ζ and Cauchy's integral formula evaluate the integral as

$$\frac{F(t, p)}{(t - \alpha)(t - \beta)} + \frac{F(\alpha, p)}{(\alpha - t)(\alpha - \beta)} + \frac{F(\beta, p)}{(\beta - t)(\beta - \alpha)}.$$

Rearranging gives RUC5 with the polynomial RUC4. The contour denominators stay nonzero on $|\zeta| = R$, so the integral is holomorphic in t, α, β and the parameters, including their coincidences. The divided-difference integral gives the same continuation of L . Taking limits therefore proves RUC5 also at coincidences, without dividing by a vanishing separation. Equivalently the integral is the actual quotient $(F - L)/((t - \alpha)(t - \beta))$ at its removable points.

The principal part itself is, exactly,

$$\begin{cases} \frac{F(\alpha, p)}{\alpha - \beta} \frac{1}{t - \alpha} + \frac{F(\beta, p)}{\beta - \alpha} \frac{1}{t - \beta}, & \alpha \neq \beta, \\ \frac{F(a, p)}{(t - a)^2} + \frac{\partial_t F(a, p)}{t - a}, & \alpha = \beta = a. \end{cases} \quad (\text{RUC6})$$

The double-pole term cannot lose its nonzero coefficient, because $F(a, p) \neq 0$. Its simple-pole coefficient is the actual derivative of the full unit reciprocal, not a coefficient that may be omitted.

Write $H(t, p) = \sum_{n \geq 0} r_n(p) t^n$. Expanding $1/(\zeta - t)$ in its uniformly convergent geometric series on any $|t| \leq R' < R$, and integrating each term, proves

$$r_n(p) = \frac{1}{2\pi i} \int_{|\zeta|=R} \frac{F(\zeta, p)}{(\zeta - \alpha)(\zeta - \beta)\zeta^{n+1}} d\zeta, \quad |r_n(p)| \leq CR^{-n} \quad (n \geq 0, p \in \mathcal{K}). \quad (\text{RUC7})$$

This estimate has one fixed radius and one fixed constant for every coefficient and every parameter in the neighborhood. It is stronger in its quantifiers than a remainder for one fixed coefficient as the parameter tends to the base point.

4.3 Every original inverse coefficient, including root collision

For the old fixed domain the inverse exists exactly off $\alpha\beta = 0$. At such a parameter let $h_n(p) = [t^n]g(t, p)^{-1}$. The complete formula is

$$\begin{aligned} h_n(p) &= P_n(p) + r_n(p), \\ P_n(p) &= \begin{cases} \frac{F(\beta, p)\beta^{-n-1} - F(\alpha, p)\alpha^{-n-1}}{\alpha - \beta}, & \alpha \neq \beta, \\ (n+1)F(a, p)a^{-n-2} - \partial_t F(a, p)a^{-n-1}, & \alpha = \beta = a \neq 0. \end{cases} \end{aligned} \quad (\text{RUC8})$$

Geometric expansion of each simple pole in RUC6 proves the first case. Differentiation of $(t-a)^{-1}$ with respect to a gives the coefficient $(n+1)a^{-n-2}$ of the double pole, proving the second, including both signs. The same second expression is the negative derivative of $F(a, p)a^{-n-1}$, which proves directly that it is the diagonal limit of the first expression.

Let $r = \min\{|\alpha|, |\beta|\} > 0$. Without any division by the separation of the roots, the finite geometric-product formula gives the uniform coefficient bound

$$|P_n(p)| \leq A(n+1)r^{-n-2}, \quad |h_n(p)| \leq A(n+1)r^{-n-2} + CR^{-n}. \quad (\text{RUC9})$$

For its proof the coefficient of t^n in q^{-1} , where $q = (t - \alpha)(t - \beta)$, is $\sum_{j=0}^n \alpha^{-j-1} \beta^{-(n-j)-1}$. Its modulus is at most $(n+1)r^{-n-2}$, and the coefficient of t^{n-1} is at most nr^{-n-1} , with zero value at $n = 0$. Multiply by $\ell_0 + \ell_1 t$ and use RUC4's bounds, $r \leq r_0$, and the definition of A . This proof works unchanged at equal roots and retains the full unit in the interpolation coefficients.

The following exact adjacent identities are valid for every $n \geq 1$:

$$\begin{aligned}\alpha P_n - P_{n-1} &= F(\beta, p) \beta^{-n-1}, \\ \beta P_n - P_{n-1} &= F(\alpha, p) \alpha^{-n-1}.\end{aligned}\tag{RUC10}$$

They follow by substituting the first formula in RUC8 and cancelling its two terms. Substitution of the second formula gives the same identities at equal roots, with the unit derivative terms cancelling exactly. Thus no two adjacent principal coefficients vanish at a parameter where the old inverse exists. For the exact coefficients, RUC7 gives, for example,

$$|\alpha h_n - h_{n-1} - F(\beta, p) \beta^{-n-1}| \leq C(|\alpha| + R) R^{-n}.\tag{RUC11}$$

The other identity has the corresponding β in this bound. Select the one whose right-hand root has modulus r . Its left-hand linear combination has modulus at most $(1 + r_0) \max\{|h_n|, |h_{n-1}|\}$. It follows, uniformly throughout the punctured neighborhood, that

$$\max\{|h_n|, |h_{n-1}|\} \geq \frac{mr^{-n-1} - C(r_0 + R)R^{-n}}{1 + r_0}.\tag{RUC12}$$

In particular define the explicit integer

$$n_1 = \max \left\{ 1, \left\lceil \frac{\log^+(2C(r_0 + R)r_0/m)}{\log(R/r_0)} \right\rceil \right\}, \quad \log^+ y = \max\{0, \log y\}.$$

For all $n \geq n_1$ the error term in RUC12 is at most half of its first term, because its relative bound is $C(r_0 + R)r_0(r_0/R)^n/m$. Consequently

$$\max\{|h_n(p)|, |h_{n-1}(p)|\} \geq \frac{m}{2(1 + r_0)} r^{-n-1} \quad (n \geq n_1).\tag{RUC13}$$

There is no exclusion of small root separation or of phase cancellation in this estimate.

4.4 The exact real phase cancellations and their limits

For a real parameter, RLA10 proves $\beta = -\bar{\alpha}$ and $F(-\bar{t}, p) = \overline{F(t, p)}$. If the roots are distinct and nonzero, write their actual values $\alpha = re^{i\phi}$ and $F(\alpha, p) = |F(\alpha, p)|e^{i\psi}$; here $\cos \phi \neq 0$, and $\beta = -re^{-i\phi}$. Then RUC8 becomes the exact parity formula

$$P_n = \begin{cases} -\frac{|F(\alpha, p)|}{r^{n+2} \cos \phi} \cos(\psi - (n+1)\phi), & n \text{ even}, \\ -\frac{i|F(\alpha, p)|}{r^{n+2} \cos \phi} \sin(\psi - (n+1)\phi), & n \text{ odd}. \end{cases}\tag{RUC14}$$

Both the numerator amplitude and phase are those of the full actual F . Thus the exact criterion for cancellation of the two root contributions is

$$P_n = 0 \iff \left(\frac{\alpha}{\beta}\right)^{n+1} = \frac{F(\alpha, p)}{F(\bar{\alpha}, p)}.\tag{RUC15}$$

This follows also directly from equating the two terms in RUC8. If α/β has infinite multiplicative order, at most one nonnegative integer n can satisfy it: two solutions would give a nonzero integer power of this ratio equal to one. If its order is the integer $q \geq 2$, either there is no solution

or the solutions are exactly one residue class for $n + 1$ modulo q . Indeed multiplication by a fixed solution reduces the equation to $(\alpha/\beta)^{n-n_0} = 1$. The ratio cannot have order one in the distinct-root case. These assertions classify actual phase cancellation; they do not assert that an independently prescribed value of the unit phase is realized.

If the two real-slice roots coincide, $\alpha = \beta = a \neq 0$ is purely imaginary. Reflection gives $F(a, p) \in \mathbb{R} \setminus \{0\}$ and $\partial_t F(a, p) \in i\mathbb{R}$. Formula RUC8 gives

$$P_n = F(a, p)a^{-n-2} \left(n + 1 - \frac{a\partial_t F(a, p)}{F(a, p)} \right). \quad (\text{RUC16})$$

The real constant in the last quotient can equal $n + 1$ for at most one nonnegative integer. Hence the double-pole case also has at most one cancelled principal coefficient. Whenever a principal coefficient cancels, the complete coefficient equals the analytic remainder and satisfies $|h_n| \leq CR^{-n}$. It need not be zero. Conversely RUC13 proves that such cancellation cannot persist in two adjacent full coefficients at the exponential scale of the actual smaller root. This is a proved limitation on cancellation, not an assumption of generic phase.

4.5 A second-order determinant identity and its explicit error

For every $n \geq 0$ and every parameter off the zero branches,

$$P_n P_{n+2} - P_{n+1}^2 = -\frac{F(\alpha, p)F(\beta, p)}{(\alpha\beta)^{n+3}}. \quad (\text{RUC17})$$

For distinct roots write $P_n = A_\alpha \alpha^{-n} + A_\beta \beta^{-n}$, where $A_\alpha = -F(\alpha, p)/(\alpha(\alpha - \beta))$ and $A_\beta = F(\beta, p)/(\beta(\alpha - \beta))$. Expanding the determinant cancels both pure square terms and leaves $A_\alpha A_\beta (\alpha\beta)^{-n} (\alpha^{-1} - \beta^{-1})^2$, which is the displayed expression. The diagonal limit exists by RUC8, giving the same identity at equal nonzero roots; direct substitution of its affine factor in n gives it as well.

Let $\mathcal{E}_n$ denote

$$\mathcal{E}_n = C(|P_n|R^{-n-2} + 2|P_{n+1}|R^{-n-1} + |P_{n+2}|R^{-n}) + 2C^2 R^{-2n-2}. \quad (\text{RUC18})$$

Expansion of $h_j = P_j + r_j$ and the triangle inequality prove

$$\left| h_n h_{n+2} - h_{n+1}^2 + \frac{F(\alpha, p)F(\beta, p)}{(\alpha\beta)^{n+3}} \right| \leq \mathcal{E}_n. \quad (\text{RUC19})$$

This is valid for all n and every such complex parameter, including collision of the two nonzero roots. The stated error uses actual P_j , so no root separation factor has been discarded or hidden.

On the real slice both root moduli equal r . Put $q_0 = r_0/R = 1/4$, $a_0 = \log(R/r_0) > 0$ and

$$Q = \frac{4CAr_0^2}{m^2} \left(3 + \frac{2}{a_0 \exp(1)} \right) + \frac{2C^2 r_0^4 q_0^2}{m^2}, \quad (\text{RUC20})$$

$$n_2 = \max \left\{ 0, \left\lceil \frac{2 \log^+(2Q)}{a_0} \right\rceil \right\}.$$

RUC9 bounds the error relative to the modulus of the principal determinant by

$$\frac{\mathcal{E}_n}{|F(\alpha, p)|^2 r^{-2n-6}} \leq \frac{4CAr_0^2}{m^2} (n+3)q_0^n + \frac{2C^2 r_0^4}{m^2} q_0^{2n+2} \leq Q e^{-a_0 n/2}.$$

For the first inequality, the three mixed errors have respective factors $(n+1)r^2(r/R)^{n+2}$, $2(n+2)r^2(r/R)^{n+1}$ and $(n+3)r^2(r/R)^n$ after division; the quadratic error has factor $2C^2 r^4(r/R)^{2n+2}$. Replace r by its upper bound r_0 and use $|F| \geq m$. For the second inequality, $(n+3)e^{-a_0 n/2} \leq$

$3 + 2/(a_0 \exp(1))$ follows by maximizing $xe^{-a_0 x/2}$ on $x \geq 0$; the other exponential decreases at least as fast. The definition of n_2 now proves

$$|h_n h_{n+2} - h_{n+1}^2| \geq \frac{1}{2} |F(\alpha, p)|^2 r^{-2n-6} \geq \frac{1}{2} m^2 r^{-2n-6} \quad (n \geq n_2). \quad (\text{RUC21})$$

The same n_2 works for every nonzero real parameter in the fixed neighborhood, even as it approaches either root collision or the certified point. It is an integer independent of the cutoff.

4.6 The finite weighted consequences, with all original constants

For $D \geq 0$, let $\rho_j = \sqrt{j!/(s/2)_j}$ with $s \in \{1, a_{\text{amp}}\}$ and $a_{\text{amp}} = 4\ell + 1 \geq 9$. The original mass $(2\pi)^{s/2}$ and centers $a_{\text{amp}}/2$, $a_{\text{amp}}/2 - 4$ are those of WCF1–6 [11]; no measure is changed. In its orthonormal frames, the old inverse is exactly $B_{D,s}^{-1} = \text{diag}(\rho)^{-1} T_D(h) \text{diag}(\rho)$. An upper diagonal of gap n has norm

$$w_{D,n,s}^+ = \max_{0 \leq i \leq D-n} \frac{\rho_{i+n}}{\rho_i} = \begin{cases} \rho_n, & s = 1, \\ \rho_D / \rho_{D-n}, & s = a_{\text{amp}}, \end{cases} \quad 0 \leq n \leq D. \quad (\text{RUC22})$$

Its nonzero columns have distinct orthogonal images, so its norm is the maximum displayed ratio. Squaring the ratio gives the product of $(i+j+1)/(i+j+s/2)$ for $0 \leq j < n$. Each factor decreases with i at $s = 1$ and increases with i at $s = a_{\text{amp}}$, proving the endpoint formulas without a replacement of the weights.

Define

$$\begin{aligned} \mathcal{B}_{D,s}(p) &= \sum_{n=0}^D w_{D,n,s}^+ \{A(n+1)r^{-n-2} + CR^{-n}\}, \\ \mathcal{R}_{D,s} &= C \sum_{n=0}^D w_{D,n,s}^+ R^{-n}, \quad \underline{\rho}_{D,s} = \min\{\rho_D, \rho_{D-1}\} \quad (D \geq 1). \end{aligned} \quad (\text{RUC23})$$

The triangle inequality for the exact weighted upper diagonals and RUC9 prove $\|B_{D,s}^{-1}\| \leq \mathcal{B}_{D,s}(p)$ and that the analytic-remainder matrix has norm at most $\mathcal{R}_{D,s}$, for every D . The entries in row zero and columns $D, D-1$, together with RUC13, give the uniform lower bound

$$\|B_{D,s}^{-1}\| \geq \frac{m \underline{\rho}_{D,s}}{2(1+r_0)\rho_0} r^{-D-1} \quad (D \geq n_1). \quad (\text{RUC24})$$

Consequently, for each fixed nonzero parameter and fixed original s , phase cancellation cannot remove exponential growth with the cutoff: in fact $\lim_{D \rightarrow \infty} \|B_{D,s}^{-1}\|^{1/D} = r^{-1} > 1$. For completeness the subexponential nature of the fixed- s weights follows directly from $\log \rho_D = -\frac{1}{2} \sum_{j=1}^D \log(1 + (s/2 - 1)/j)$. After finitely many terms, the absolute logarithm of each factor is at most $2|s/2 - 1|/j$, by integrating $1/(1+t)$ between zero and that number. Hence $|\log \rho_D| = O_s(1 + \log D)$. The finite sums in RUC23 then have the same D th-root upper rate r^{-1} , since $r < R < 1$ and their remaining factors are polynomial in D for fixed s . The lower bound gives the matching rate. If s changes with D , RUC22–24 remain the exact bounds; no subexponential assertion about that change is inferred from this fixed- s argument.

On the real slice and for $D \geq n_2 + 2$, the minor in rows $0, 1$ and columns $D-1, D$ of the exact inverse has determinant

$$\frac{\rho_{D-1} \rho_D}{\rho_0 \rho_1} (h_{D-1}^2 - h_D h_{D-2}).$$

The exterior norm dominates the absolute value of each matrix entry in orthonormal exterior frames. RUC21 therefore proves

$$\left\| \bigwedge^2 B_{D,s}^{-1} \right\| \geq \frac{m^2 \rho_{D-1} \rho_D}{2\rho_0 \rho_1} r^{-2D-2}. \quad (\text{RUC25})$$

For fixed s , RUC23 and the inequality $\|\bigwedge^2 B^{-1}\| \leq \|B^{-1}\|^2$ give the exact D th-root rate r^{-2} on this real slice, including every phase-cancelling subsequence and the double-root case. For every exterior rank $1 \leq k \leq D+1$, the complete finite upper bound $\|\bigwedge^k B^{-1}\| \leq \mathcal{B}_{D,s}^k$ also follows. At the top rank its exact value is $|g_0(p)|^{-D-1}$, because the diagonal of B is g_0 . The complementary exterior identity $\|\bigwedge^k B^{-1}\| = \|\bigwedge^{D+1-k} B\|/|g_0|^{D+1}$ retains every original forward coefficient. These finite formulas do not assert an all-rank asymptotic obtained by discarding the analytic remainder.

4.7 The full-matrix pole terms and their lower-triangular correction

Formula RUC8 for P_n extends to every integer n when the roots are nonzero. The full matrix $(P_{j-i}\rho_j/\rho_i)_{0 \leq i,j \leq D}$ has rank at most two. For distinct roots it is exactly the sum of the two outer products

$$A_\alpha(\alpha^i/\rho_i)_i(\rho_j\alpha^{-j})_j^\top + A_\beta(\beta^i/\rho_i)_i(\rho_j\beta^{-j})_j^\top, \quad (\text{RUC26})$$

with the actual amplitudes in the proof of RUC17. At equal roots the expression RUC8 is affine in $j-i$ times a^{i-j} , so its columns lie in the span of $(a^i/\rho_i)_i$ and $(ia^i/\rho_i)_i$; this proves rank at most two without taking an unbounded separated-pole representation. The rank assertion does not apply directly to its upper-triangular part. The difference is a required lower-triangular correction, whose complete entries at gap $l = i - j \geq 1$ are $-P_{-l}\rho_j/\rho_i$. They have the exact coefficient formula and uniform bound

$$P_{-l} = - \int_0^1 \partial_z \{F(z, p) z^{l-1}\} \Big|_{z=\beta+\theta(\alpha-\beta)} d\theta, \quad (\text{RUC27})$$

$$|P_{-l}| \leq C_1 r_0^{l-1} + (l-1)C_0 r_0^{l-2}.$$

For $l = 1$ the second term is defined to be zero. The first formula is precisely the negative divided difference of $F(z, p) z^{l-1}$, and includes its derivative at equal roots. The segment bound follows by differentiation and RUC3; the polynomial makes this integral regular even if the segment passes through zero. The exact norm of the lower shift weight is $\max_{0 \leq j \leq D-l} \rho_j/\rho_{j+l}$, equal to ρ_{D-l}/ρ_D at $s = 1$ and ρ_0/ρ_l at $s = a_{\text{amp}}$, by the same product argument as RUC22. Thus its full finite correction norm is at most the sum of these weights times the right side of RUC27 for $1 \leq l \leq D$. Together with $\mathcal{R}_{D,s}$ this is an exact rank-two-plus- remainder decomposition of the weighted inverse, retaining every lower shift. It supplies finite constants for further exterior estimates rather than replacing an upper-triangular pole matrix by a rank-two matrix without its correction. To record its full exterior consequence define

$$w_{D,l,s}^- = \max_{0 \leq j \leq D-l} \frac{\rho_j}{\rho_{j+l}},$$

$$\mathcal{E}_{D,s}^{\text{mat}} = \mathcal{R}_{D,s} + \sum_{l=1}^D w_{D,l,s}^- \{C_1 r_0^{l-1} + (l-1)C_0 r_0^{l-2}\}.$$

As before the second summand inside braces is zero at $l = 1$. For every $1 \leq k \leq D+1$, at every off-branch parameter in the fixed neighborhood, one has the complete finite bound

$$\|\bigwedge^k B_{D,s}^{-1}\| \leq \mathcal{B}_{D,s}(p)^{\min(k,2)} \min\{\mathcal{B}_{D,s}(p), \mathcal{E}_{D,s}^{\text{mat}}\}^{\max(k-2,0)}. \quad (\text{RUC28})$$

Indeed the rank-two full pole matrix has a kernel of codimension at most two. On that kernel the full inverse equals the remainder, whose norm is at most $\mathcal{E}_{D,s}^{\text{mat}}$. Diagonalizing the positive matrix of the inverse squared norm proves that its third and subsequent singular values are at most this bound: if three were larger, their three-dimensional singular-vector space would intersect that kernel nontrivially, contradicting the norm bound there. Every singular value is also at most $\mathcal{B}_{D,s}(p)$ by RUC23. Multiplying the largest k singular values proves RUC28, including dimensions one and two. The radius is fixed and smaller than one, so the analytic part $\mathcal{R}_{D,s}$ is not claimed bounded as D tends to infinity.

4.8 The actual order-one and order-two domains

At a point of either single zero branch, let $c \neq 0$ denote its remaining root; it satisfies $|c| \leq r_0$. The actual order is one and the reciprocal of the actual quotient symbol is $t/g = F(t, p)/(t - c)$. On their intersection the actual order is two and $t^2/g = F(t, p)$. Their complete coefficients are

$$\begin{array}{c|c|c} v & [t^n](t^v/g) & \text{uniform analytic coefficient error} \\ \hline 1 & -F(c, p)c^{-n-1} + s_n(p) & |s_n(p)| \leq M(R - r_0)^{-1}R^{-n} \\ 2 & [t^n]F(t, p) & |[t^n]F(t, p)| \leq MR^{-n}. \end{array} \quad (\text{RUC29})$$

For $v = 1$, divide $F(t, p) - F(c, p)$ by $t - c$ using $\frac{1}{2\pi i} \int_{|\zeta|=R} F(\zeta, p)/((\zeta - c)(\zeta - t)) d\zeta$. The two-point partial fraction identity and Cauchy's formula prove the division, and coefficient extraction gives the stated bound. The $v = 2$ case is Cauchy's coefficient integral itself. The original weighted inverse on these actual strata has entries

$$(\mathcal{A}_{D,s,p}^{-1})_{ij} = [t^{j-i}](t^v/g) \frac{\rho_j}{\rho_{i+v}} \quad (j \geq i), \quad (\text{RUC30})$$

with zero entries otherwise, by multiplying the weighted reciprocal from WCF6 and RLA30. Its source and target remain the original $\mathcal{P}_{D+v}/\mathcal{P}_{v-1}$ and $\mathcal{P}_D$, with the unchanged Gamma norms. Summing the exact diagonal weight maxima against RUC29 gives finite norm and exterior bounds on these domains as well. The old inverse has no Taylor coefficient at zero-root parameters and has not been extended there. All strata are connected by the original maps and the complete unit, not by replacing their domains or leading moments.

5 Both inverse singular rates at the original growing cutoffs

This section uses the exact factorization RLA6 [7] and the full original Gamma frames of WCF1–6 [11, 3]. The complete RLA source accompanies this continuation; WCF has its pinned original proof. The general principal-part rank decomposition and its lower corrections are already WCF15–18. Here their exact two-root specialization yields both singular rates and the retained unit contribution. It calculates a limit in the cutoff with the actual period and unit held fixed. All four ES states and the original unit remain those of RLA1–35. The local parameter choices below are points of that proved geometric family; no identification with an arithmetic zeta quartet is made.

Choose fixed radii $0 < r_0 < R < 1$ and a sufficiently small closed parameter neighborhood of the certified point such that the full unit $U(t, p)$ in RLA6 is holomorphic and nonzero for $|t| \leq R$, and both actual roots obey $|\alpha|, |\beta| < r_0$. Such choices exist by RLA5–6 and compactness after shrinking the original analytic neighborhoods. They are independent of D . Throughout the inverse calculation fix an actual parameter in this neighborhood with $\alpha\beta \neq 0$ and put

$$F(t) = U(t, p)^{-1}, \quad r = \min(|\alpha|, |\beta|), \quad s_r = \max(|\alpha|, |\beta|), \quad M_R = \max_{|t|=R} |F(t)|. \quad (\text{RCM1})$$

The symbol s_r is a root modulus, not the original source order s . The function F here is the reciprocal conductor unit; AUH2's entire centered Xi function is a different, explicitly defined function. All constants in the following formulas refer to these exact functions, not to assigned residues or a changed symbol.

5.1 Two complete principal parts and the remaining analytic function

For distinct roots let $L(t)$ be the exact linear interpolation of F at α, β ; for equal roots let it be its first Taylor polynomial there. The identity

$$\frac{1}{g(t, p)} = \frac{L(t)}{(t - \alpha)(t - \beta)} + H(t), \quad H(t) = \frac{F(t) - L(t)}{(t - \alpha)(t - \beta)} \quad (\text{RCM2})$$

retains the entire original unit. The apparent singularities of H are removable by the interpolation identities, including the double-root case. An exact formula, including that case, is

$$H(t) = \frac{1}{2\pi i} \int_{|z|=R} \frac{F(z)}{(z-\alpha)(z-\beta)(z-t)} dz, \quad |t| < R. \quad (\text{RCM3})$$

For distinct roots this follows by adding the residues at $z = t, \alpha, \beta$: the residue at t is $F(t)/q(t)$; the other two together are $-L(t)/q(t)$. For a repeated root the residue is the derivative at that root and gives its first Taylor polynomial. The formula can also be obtained by a limit of the distinct-root formula, uniformly on a smaller circle. Taking the Taylor coefficient of the Cauchy kernel proves

$$h_n = [t^n]g^{-1} = P_n + e_n, \quad |e_n| \leq C_R R^{-n}, \quad C_R = \frac{M_R}{(R-r_0)^2}, \quad (\text{RCM4})$$

for every integer $n \geq 0$ simultaneously. For example the coefficient integral has denominator z^{n+1} ; its circle length $2\pi R$ gives exactly the displayed bound.

The full principal coefficient is

$$P_n = \begin{cases} \frac{F(\beta)\beta^{-n-1} - F(\alpha)\alpha^{-n-1}}{\alpha - \beta}, & \alpha \neq \beta, \\ (n+1)F(\alpha)\alpha^{-n-2} - F'(\alpha)\alpha^{-n-1}, & \alpha = \beta. \end{cases} \quad (\text{RCM5})$$

Geometric expansion of the simple pole terms, or of the double pole and its derivative term, proves this formula. It also defines P_n for every negative integer n , an algebraic extension that will be used for a finite matrix identity only. It does not insert negative Taylor coefficients into g^{-1} .

Write $C_0 = \max_{|t| \leq r_0} |F(t)|$ and $C_1 = \max_{|t| \leq r_0} |F'(t)|$. Cauchy's formula gives $C_0 \leq M_R R / (R - r_0)$ and $C_1 \leq M_R R / (R - r_0)^2$. For the full interpolation $L(t) = A_0 + A_1 t$, integration along the segment between the roots gives $|A_1| \leq C_1$ and $|A_0| \leq C_0 + r_0 C_1$, also for a repeated root. Expanding both quadratic factors as geometric series therefore gives the all- n bound

$$|P_n| \leq (n+1)(C_0 + 2r_0 C_1) r^{-n-2} \quad (n \geq 0). \quad (\text{RCM6})$$

Indeed the reciprocal quadratic coefficient is a sum of $n+1$ terms, each bounded by r^{-n-2} ; the linear part contributes at most $n C_1 r^{-n-1}$. These retain both original root moduli; using their minimum here is an explicitly proved upper bound.

5.2 The exact finite rank-two matrix and its full correction

Let $N = D + 1$ and $T_D(h)_{ij} = h_{j-i}$ for $j \geq i$, zero otherwise. Define $\mathcal{P}_D = (P_{j-i})_{0 \leq i, j \leq D}$ using the full integer formula RCM5. Then

$$T_D(h) = \mathcal{P}_D + \mathcal{E}_D - \mathcal{L}_D, \quad (\mathcal{E}_D)_{ij} = \mathbf{1}_{j \geq i} e_{j-i}, \quad (\mathcal{L}_D)_{ij} = \mathbf{1}_{i > j} P_{j-i}. \quad (\text{RCM7})$$

This is equality of every matrix entry. In the distinct-root case $\mathcal{P}_D$ is the sum of the two outer products

$$-\frac{F(\alpha)}{(\alpha - \beta)\alpha} (\alpha^i)_i (\alpha^{-j})_j^\top + \frac{F(\beta)}{(\alpha - \beta)\beta} (\beta^i)_i (\beta^{-j})_j^\top. \quad (\text{RCM8})$$

It has rank at most two. At coalescence RCM5 is $\alpha^{i-j-2}((j-i+1)F(\alpha) - \alpha F'(\alpha))$; grouping the terms constant and linear in i gives two outer products again. Thus the same rank bound holds at coalescence without a singular divided-difference representation.

For $k \geq 1$, apply the segment derivative formula to $F(z)z^{k-1}$ in RCM5. It proves

$$|P_{-k}| \leq C_1 r_0^{k-1} + \mathbf{1}_{k \geq 2} C_0 (k-1) r_0^{k-2}, \quad \|\mathcal{L}_D\| \leq K_- := \frac{C_1}{1-r_0} + \frac{C_0}{(1-r_0)^2}, \quad \|\mathcal{E}_D\| \leq \frac{C_R R^{-D}}{1-R}. \quad (\text{RCM9})$$

For the matrix bounds each shift along a fixed diagonal has norm at most one; sum its absolute coefficient. The geometric series and its derivative give K_- , and $\sum_{n=0}^D R^{-n} \leq R^{-D}/(1-R)$ gives the last bound. Hence every correction to the rank-two expression is explicit, with constants independent of the cutoff.

5.3 Both singular growth rates, including phase cancellations

Two identities obtained by multiplying RCM5 are

$$\begin{aligned} \alpha P_n - P_{n-1} &= F(\beta)\beta^{-n-1}, \\ \beta P_n - P_{n-1} &= F(\alpha)\alpha^{-n-1}, \quad n \geq 1, \\ P_n P_{n+2} - P_{n+1}^2 &= -\frac{F(\alpha)F(\beta)}{(\alpha\beta)^{n+3}}. \end{aligned} \tag{RCM10}$$

Substitution in RCM5 proves the first two identities term by term. For the last, write $P_n = A\alpha^{-n} + B\beta^{-n}$ as in RCM8. The two square terms cancel; the remaining term is $AB(\alpha^{-1} - \beta^{-1})^2(\alpha\beta)^{-n}$, which is the stated expression. Its limit proves the same identity for a repeated root; direct expansion gives it too.

Since F has no zero, set $m_F = \min(|F(\alpha)|, |F(\beta)|) > 0$. Choose the second or first identity in RCM10 according to which root has modulus r . RCM4 gives, for every $n \geq 1$,

$$(1 + r_0) \max(|h_n|, |h_{n-1}|) \geq m_F r^{-n-1} - C_R(r_0 + R)R^{-n}. \tag{RCM11}$$

The error-to-main ratio is at most $C_R(r_0 + R)r_0(r_0/R)^n/m_F$, which tends to zero independently of D . In particular half the first term remains for every n beyond the explicit threshold

$$n_0 = \max \left\{ 1, \left\lceil \frac{\max\{0, \log(2C_R(r_0 + R)r_0/m_F)\}}{\log(R/r_0)} \right\rceil \right\}. \tag{RCM12}$$

Thus cancellations of a single coefficient cannot remove the exponential growth at arbitrarily large cutoffs. Every coefficient appears as an entry in the upper triangular matrix. RCM6 and RCM4 give an upper bound with only a quadratic factor in D times r^{-D-2} . RCM11 with $n = D$ gives the matching lower exponential rate, so for its largest singular value

$$\lim_{D \rightarrow \infty} \sigma_1(T_D(h))^{1/D} = r^{-1}. \tag{RCM13}$$

For the second singular value, the corner minor with rows $0, 1$ and columns $D-1, D$ is $h_{D-1}^2 - h_D h_{D-2}$. RCM10 and the remainder bounds show

$$h_{D-1}^2 - h_D h_{D-2} = \frac{F(\alpha)F(\beta)}{(\alpha\beta)^{D+1}}(1 + o(1)) \quad (D \rightarrow \infty). \tag{RCM14}$$

Here the point is fixed, including its nonzero root moduli. To verify the relative error, each cross term is bounded using RCM6 and RCM4 by a fixed constant times $(D+1)r^{-D-2}R^{-D+2}$; the two remainder products have bound $2C_R^2 R^{-2D+2}$. Dividing by the nonzero principal term makes these bounds at most fixed constants times $(D+1)(s_r/R)^D$ and $(rs_r/R^2)^D$, respectively. Both tend to zero because $s_r < R$. The constants may depend on the fixed roots; no uniform claim at $r = 0$ is inferred. More explicitly, put $A = C_0 + 2r_0 C_1$ and $J = |F(\alpha)F(\beta)|$. The relative error is bounded by

$$\frac{4AC_R R^2 s_r}{rJ} (D+1)(s_r/R)^D + \frac{2C_R^2 R^2 r s_r}{J} (rs_r/R^2)^D.$$

Indeed the three mixed terms are bounded by $AC_R r^{-D-2} R^{-D} \{2DrR + (D+1)R^2 + (D-1)r^2\}$; use $r < R$ and divide by $J/(rs_r)^{D+1}$. The two remaining products have sum at most $2C_R^2 R^{-2D+2}$.

The modulus of this minor is at most $\sigma_1(T_D(h))\sigma_2(T_D(h))$, since it is an entry of the second exterior matrix. RCM13–14 therefore prove $\liminf \sigma_2(T_D(h))^{1/D} \geq s_r^{-1}$. For the upper bound, if $r < s_r$, subtract the rank-one term in RCM8 belonging to the smaller root. The other term has norm bounded by a constant times $(D+1)s_r^{-D}$, and RCM9 has strictly smaller exponential rate $R^{-1} < s_r^{-1}$. The minimum–maximum characterization of singular values bounds σ_2 by the norm of that residual: restrict to the kernel of the subtracted rank-one map. When $r = s_r$, simply use $\sigma_2 \leq \sigma_1$ and RCM13. This also covers coalescence. We have proved

$$\begin{aligned} \lim_{D \rightarrow \infty} \sigma_2(T_D(h))^{1/D} &= s_r^{-1}, \\ \lim_{D \rightarrow \infty} \left\| \bigwedge^2 T_D(h) \right\|^{1/D} &= |\alpha\beta|^{-1}, \\ \limsup_{D \rightarrow \infty} \sigma_3(T_D(h))^{1/D} &\leq R^{-1}. \end{aligned} \tag{RCM15}$$

For the third bound, restrict to the kernel of the rank-at-most-two matrix in RCM7 and use RCM9. This proof applies for $D \geq 2$, which suffices for the stated limit.

5.4 The original amplification and the retained unit contribution

The original inverse in its original orthonormal Gamma frames is $B_{D,s}^{-1} = D_\rho^{-1} T_D(h) D_\rho$. The full source and target masses and centers remain those of RLA21; both coefficient frames are orthonormal for those measures. Define $\kappa_{D,s} = \max_{j \leq D} \rho_j / \min_{j \leq D} \rho_j$. For every singular index, multiplication by this invertible diagonal matrix gives

$$\kappa_{D,s}^{-1} \sigma_j(T_D(h)) \leq \sigma_j(B_{D,s}^{-1}) \leq \kappa_{D,s} \sigma_j(T_D(h)), \quad \begin{cases} \log \kappa_{D,1} \leq \frac{1}{4} \log(4D+1), \\ \log \kappa_{D,s} \leq \frac{1}{2}(s/2-1)H_D, \quad s \geq 2. \end{cases} \tag{RCM16}$$

Here $H_D = \sum_{j=1}^D 1/j$, unrelated to RCM2’s analytic function $H(t)$. The singular inequalities follow from the minimum–maximum principle applied to both diagonal factors and then to their inverses. For $s = 1$, the original weight induction in RQT15 gives the first bound. For $s \geq 2$, ρ_j decreases, and $\kappa_{D,s}^2 = \prod_{j=1}^D (1 + (s/2-1)/j)$; use $\log(1+x) \leq x$ termwise. No weight is deleted.

Consequently all three singular-rate statements RCM13 and RCM15 hold for the actual weighted inverse whenever $s = 1$, fixed s , or $s = s_D \geq 2$ with $s_D \log(D+1)/D \rightarrow 0$. In particular they hold for the original amplification $s_D = a_{\text{amp},D} = 4\ell_D + 1 \geq 9$ on every sequence of the original cutoffs with $a_{\text{amp},D} = O(\sqrt{D})$. In particular WCF1 and WCF25 give $q = (a_{\text{amp}} + 1)^2$ and $D_i = (q-1-v, q-v, 2q-1-v, 2q-v)_i$; here the old invertible family has $v = 0$. Each of these four original endpoint sequences has $D_i = \Theta(a_{\text{amp}}^2)$, so it satisfies the stated condition. The proof is that $\kappa_{D,s_D}^{1/D} \rightarrow 1$ in RCM16; for the second exterior power the multiplicative factor is at most κ_{D,s_D}^2 , with the same conclusion. This proves uniformity in the growing matrix size at each fixed admitted geometric parameter, rather than applying a fixed- D local Taylor expansion at changing cutoffs.

There is also an exact evaluation of the remaining total logarithm. Let the inverse singular values be $\tau_{1,D} \geq \dots \geq \tau_{D+1,D} > 0$ in any of the original source sequences just specified. Triangularity and the exact factorization give

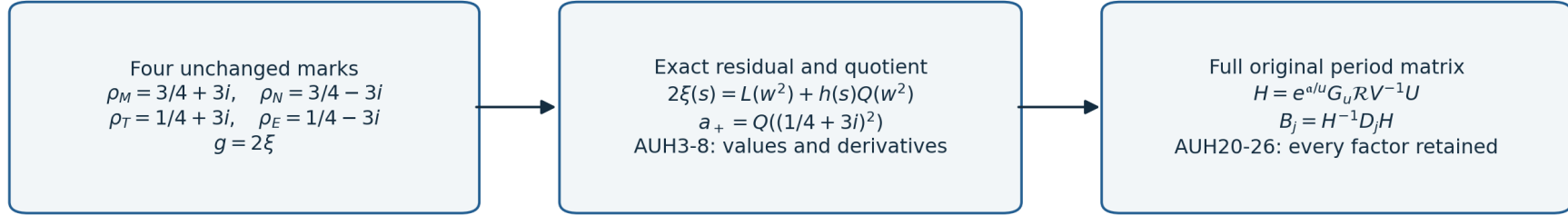
$$\prod_{j=1}^{D+1} \tau_{j,D} = |g_0|^{-D-1}, \quad g_0 = U(0,p)\alpha\beta, \quad \lim_{D \rightarrow \infty} \frac{1}{D} \sum_{j=3}^{D+1} \log \tau_{j,D} = -\log |U(0,p)|. \tag{RCM17}$$

Take logarithms of the first identity, subtract the first two singular logarithms and use RCM13 and RCM15. The root factors cancel exactly, leaving the original analytic unit. Its value is not set to one, and its contribution is extensive in the dimension. This identity does not evaluate

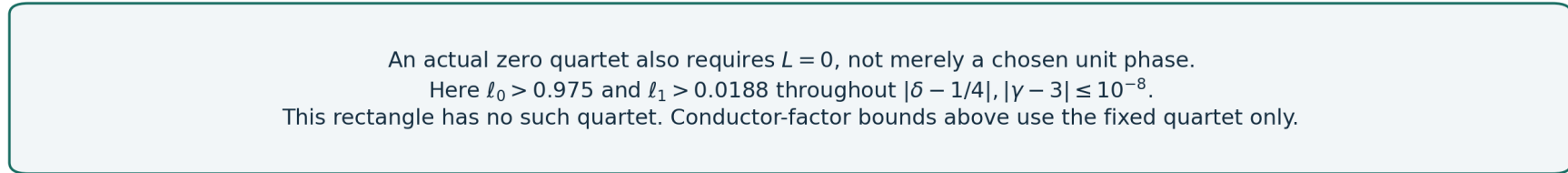
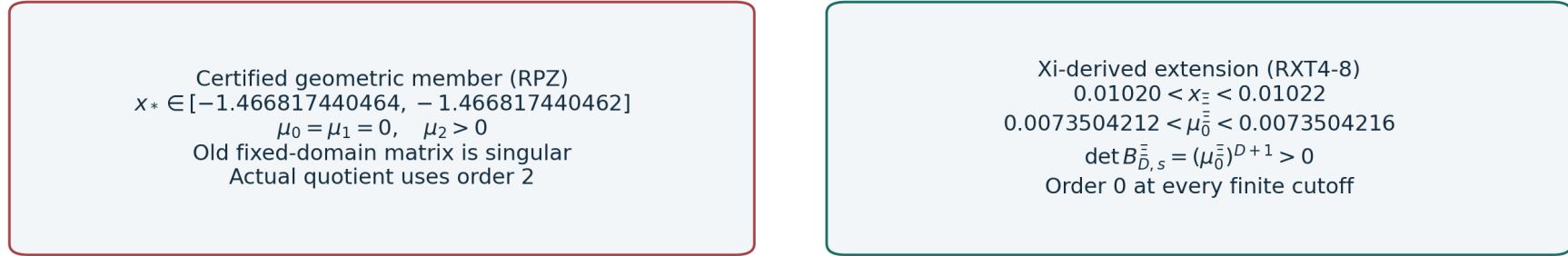
arbitrary intermediate exterior ranks growing with D ; it evaluates precisely the two leading rates and the complete remaining logarithmic sum.

On the actual real parameter slice, RLA10 gives $\beta = -\bar{\alpha}$, so both leading inverse singular rates are $|\alpha|^{-1}$ and the second exterior rate is $|\alpha|^{-2}$. These statements include every phase cancellation and the repeated-root line away from zero. They concern $D \rightarrow \infty$ at a fixed invertible parameter. RPD and RQT20–22 concern approach to the singular parameter at a fixed cutoff and retain their different full constants; no interchange of these limits has been made.

The actual Xi data decide which unit enters the conductor



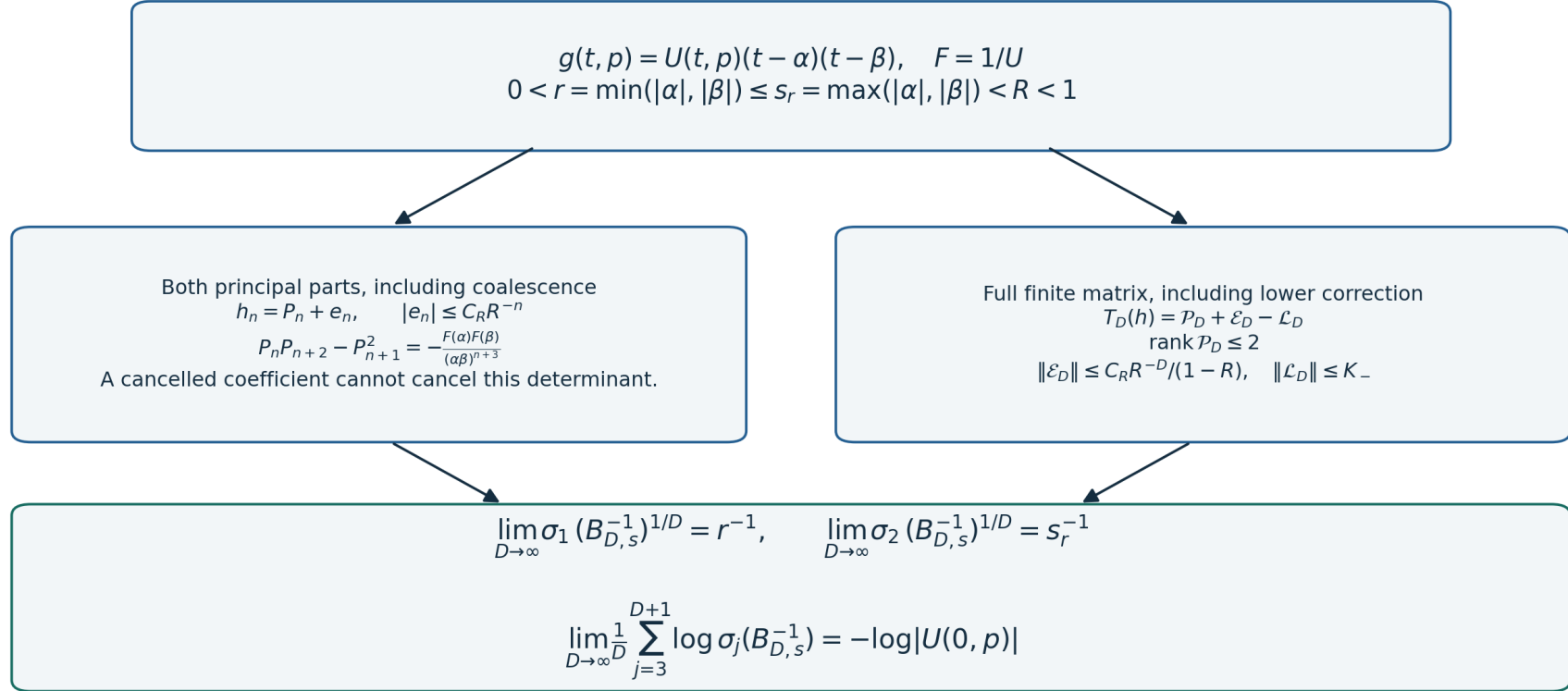
Same quartet and same certified real period interval; two different unit inputs



Origin: Levent Alpöge's announcement, crediting Akhil and Fable; four-mark extension: ES reader.
 Exact maps: ALF1-3, AUH1-30, RXT1-9; original family: PCL1-10.
 Riemann completion: Apostol, DLMF Ch. 25; interval arithmetic: Johansson, Arb, arXiv:1611.02831v1.
 Schematic box positions. Full primary-source links and proof locators accompany this figure.

Figure 1: The exact Xi-jet morphism carries the residual L as well as the quotient values. AUH1–30 proves the maps. RXT1–9 certifies a positive zeroth moment for the extension over the full original period interval, and a separate residual exclusion on the stated quartet rectangle. The latter does not enlarge the domain of the conductor-factor bounds. The geometric singular point remains the one proved in [8]; the original period family is [6]. Riemann completion and interval arithmetic: [4, 12]. The originating Jacobian counterexample is Alpöge–Fable [1]; the four-mark extension is [5]. Box positions are schematic.

Two spectral roots control two leading inverse rates



The parameter p is fixed and invertible. Both roots may coincide away from zero.
 Original Gamma weights stay: $B^{-1} = D_p^{-1} T_D(h) D_p$; $s_D \log(D+1)/D \rightarrow 0$ covers all four original $D_i = \Theta(a_{\text{amp}}^2)$ endpoints.

Origin: Levent Alpöge's announcement, crediting Akhil and Fable; four-mark extension: ES reader.
 RUC1-30 and RCM1-17; exact local factorization RLA6-10; preceding principal-part mechanism WCF15-18.
 All corrections and unit terms are retained. Full primary-source links accompany this figure.

Figure 2: RUC1–30 and RCM1–17 retain both actual spectral roots, the full unit, the analytic remainder and every lower-diagonal correction. The determinant identity prevents coefficient-phase cancellation from removing both leading inverse rates. The Gamma comparison applies at the four original quadratic cutoffs with the parameter fixed. The principal-part rank mechanism is the earlier WCF15–18 [11]; the full local factorization is RLA6–10 [7]. The originating collision construction is Alpöge–Fable [1], with the ES four-mark extension [5]. No numerical root neighborhood is asserted by this diagram. All constants and quantified neighborhoods are in the proofs.

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